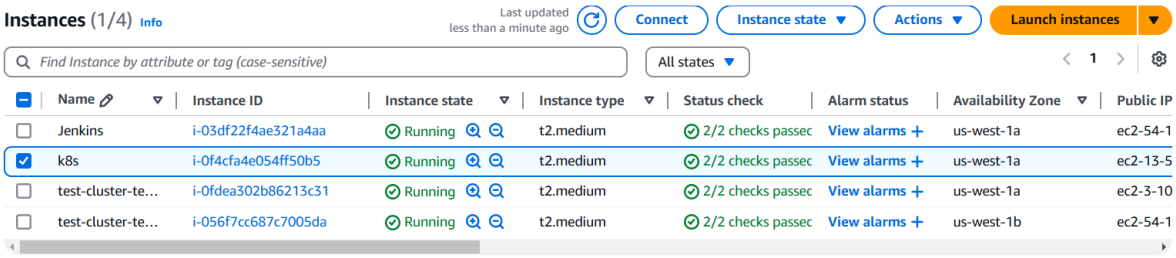
How to Set Up Eks cluster

Eks cluster

- Create a Linux server with ubuntu os for setting up eks cluster



- Connect your ec2 instance with putty or mobaxterm using shh client
- After that you have to install following cli tools

1. **Eksctl**

Description: - eksctl is a simple CLI tool for creating and managing clusters on EKS - Amazon's managed Kubernetes service for EC2. It is written in Go, uses CloudFormation

Command: # for ARM systems, set ARCH to: `arm64`, `armv6` or `armv7`

```
ARCH=amd64

PLATFORM=$(uname -s)_$ARCH


curl -sLO "https://github.com/eksctl-io/eksctl/releases/latest/download/eksctl_$PLATFORM.tar.gz"


# (Optional) Verify checksum

curl -sL "https://github.com/eksctl-io/eksctl/releases/latest/download/eksctl_checksums.txt" | grep $PLATFORM | sha256sum --check
```

```
tar -xzf eksctl_$PLATFORM.tar.gz -C /tmp && rm eksctl_$PLATFORM.tar.gz

sudo mv /tmp/eksctl /usr/local/bin
```

2. **Kubectl**

Description: - kubectl is the command-line tool for interacting with Kubernetes clusters. It allows users to manage Kubernetes resources and perform various tasks such as deploying applications, inspecting and modifying cluster resources, and troubleshooting cluster issues.

Command: -

- Install kubectl binary with curl on Linux

```
curl -LO https://dl.k8s.io/release/$(curl -L -s https://dl.k8s.io/release/stable.txt)/bin/linux/amd64/kubectl
```

- Validate the binary (optional)

Download the kubectl checksum file:

```
curl -LO https://dl.k8s.io/release/$(curl -L -s https://dl.k8s.io/release/stable.txt)/bin/linux/amd64/kubectl.sha256
```

- Validate the kubectl binary against the checksum file:

```
echo "$(cat kubectl.sha256)  kubectl" | sha256sum –check
```

- Install kubectl

```
sudo install -o root -g root -m 0755 kubectl /usr/local/bin/kubectl
```

- Test to ensure the version you installed is up-to-date:

```
kubectl version --client
```

- If kubectl version is not visible execute following commands (its optional)

```
chmod +x kubectl
```

mkdir -p ~/.local/bin

mv ./kubectl ~/.local/bin/kubectl

kubectl version --client

3. **AWS CLI**

Description: - The **AWS Command Line Interface (CLI)** is a tool provided by Amazon Web Services that enables users to interact with AWS services from a terminal or command prompt. It allows you to perform various AWS tasks directly from the command line, including managing resources, configuring settings, and automating workflows.

Command: -

apt install unzip -y

curl "https://awscli.amazonaws.com/awscli-exe-linux-x86_64.zip" -o "awscliv2.zip"

unzip awscliv2.zip

sudo ./aws/install

Confirm the installation with the following command: -

aws --version

4. **AWS access key and secret key** you have to provide before that you have to create one IAM user with administrator access permission then generate access key and secret key and you configure these keys on your terminal

aws configure (execute this command on your terminal)

```
root@k8s:/home/ubuntu# aws configure
AWS Access Key ID [****************EKUB]: AKIAST6S6Z6L6OSIEKUB
AWS Secret Access Key [****************ZZ/5]: hWsQYQLff4dVYznUqFBK1LgRE7waYazw3L+HZZ/5
Default region name [us-west-1]:
Default output format [None]:
root@k8s:/home/ubuntu#
```

- Now you can create eks cluster by executing following command

eksctl create cluster --name <cluster_name> --region <your_region> --zones <availability_zones> --nodegroup-name <nodegroip_name> --node-type <instance_type> --nodes <no.of.nodes>

- Now your cluster is ready as shown below image 👇

```
2024-11-17 10:04:49 [ℹ]  waiting for CloudFormation stack "eksctl-david-k8s-cluster"
2024-11-17 10:05:49 [ℹ]  waiting for CloudFormation stack "eksctl-david-k8s-cluster"
2024-11-17 10:06:49 [ℹ]  waiting for CloudFormation stack "eksctl-david-k8s-cluster"
2024-11-17 10:06:50 [!]  recommended policies were found for "vpc-cni" addon, but since OIDC is disabled on the cluster, eksctl cannot
configure the requested permissions; the recommended way to provide IAM permissions for "vpc-cni" addon is via pod identity association
s; after addon creation is completed, add all recommended policies to the config file, under `addon.PodIdentityAssociations`, and run `
eksctl update addon`
2024-11-17 10:06:50 [ℹ]  creating addon
2024-11-17 10:06:50 [ℹ]  successfully created addon
2024-11-17 10:06:51 [ℹ]  creating addon
2024-11-17 10:06:51 [ℹ]  successfully created addon
2024-11-17 10:06:52 [ℹ]  creating addon
2024-11-17 10:06:52 [ℹ]  successfully created addon
2024-11-17 10:08:52 [ℹ]  building managed nodegroup stack "eksctl-david-k8s-nodegroup-david-node"
2024-11-17 10:08:52 [ℹ]  deploying stack "eksctl-david-k8s-nodegroup-david-node"
2024-11-17 10:08:52 [ℹ]  waiting for CloudFormation stack "eksctl-david-k8s-nodegroup-david-node"
2024-11-17 10:09:22 [ℹ]  waiting for CloudFormation stack "eksctl-david-k8s-nodegroup-david-node"
2024-11-17 10:10:19 [ℹ]  waiting for CloudFormation stack "eksctl-david-k8s-nodegroup-david-node"
2024-11-17 10:12:10 [ℹ]  waiting for CloudFormation stack "eksctl-david-k8s-nodegroup-david-node"
2024-11-17 10:12:10 [ℹ]  waiting for the control plane to become ready
2024-11-17 10:12:11 [✔]  saved kubeconfig as "/root/.kube/config"
2024-11-17 10:12:11 [ℹ]  no tasks
2024-11-17 10:12:11 [✔]  all EKS cluster resources for "david-k8s" have been created
2024-11-17 10:12:11 [✔]  created 0 nodegroup(s) in cluster "david-k8s"
2024-11-17 10:12:11 [ℹ]  nodegroup "david-node" has 2 node(s)
2024-11-17 10:12:11 [ℹ]  node "ip-192-168-22-157.ec2.internal" is ready
2024-11-17 10:12:11 [ℹ]  node "ip-192-168-55-13.ec2.internal" is ready
2024-11-17 10:12:11 [ℹ]  waiting for at least 2 node(s) to become ready in "david-node"
2024-11-17 10:12:11 [ℹ]  nodegroup "david-node" has 2 node(s)
2024-11-17 10:12:11 [ℹ]  node "ip-192-168-22-157.ec2.internal" is ready
2024-11-17 10:12:11 [ℹ]  node "ip-192-168-55-13.ec2.internal" is ready
2024-11-17 10:12:11 [✔]  created 1 managed nodegroup(s) in cluster "david-k8s"
2024-11-17 10:12:12 [ℹ]  kubectl command should work with "/root/.kube/config", try 'kubectl get nodes'
2024-11-17 10:12:12 [✔]  EKS cluster "david-k8s" in "us-east-1" region is ready
root@ip-172-31-86-222:/home/ubuntu# kubectl get nodes
NAME                           STATUS   ROLES    AGE     VERSION
ip-192-168-22-157.ec2.internal   Ready    <none>   6m27s   v1.30.4-eks-a737599
ip-192-168-55-13.ec2.internal    Ready    <none>   6m20s   v1.30.4-eks-a737599
root@ip-172-31-86-222:/home/ubuntu#
```